

Customer Segmentation Report

KMeans Clustering Analysis | 2,627 Customers | 4 Segments

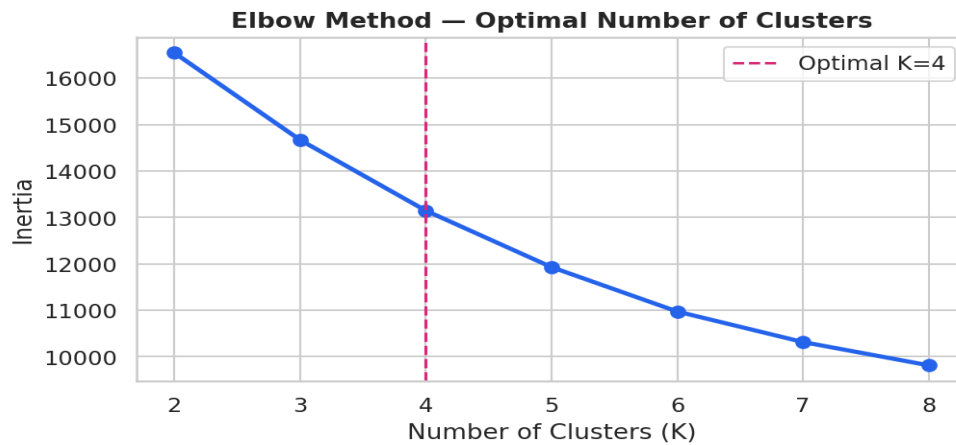
Prepared for: Alfido Tech Internship Submission
Method: KMeans Clustering (K=4) with PCA Visualization

2,627	4	Low → High	Engineer
Total Customers	Segments Found	Spending Range	Top Profession

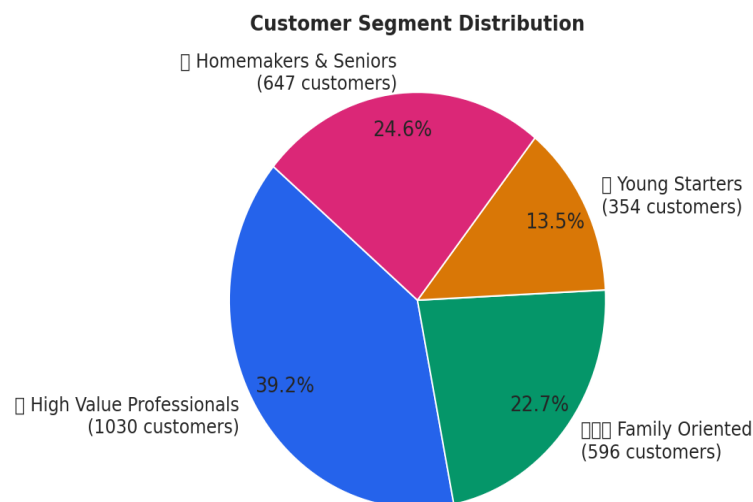
1. Methodology & Data Preparation

This dataset contains customer demographic and behavioral data — not transaction records. Since there are no purchase timestamps or monetary values, RFM analysis cannot be applied directly. Instead, KMeans Clustering was used to group customers by their profile similarities.

Step	Action	Detail
1	Load Data	2,627 rows × 10 columns, latin-1 encoding
2	Handle Missing Values	Work Experience → median fill; Profession/Married → 'Unknown'
3	Encode Categories	Gender, Married, Graduated, Profession → Label Encoded
4	Map Spending Score	Low=1, Average=2, High=3
5	Scale Features	StandardScaler — makes all features equal weight
6	Find Optimal K	Elbow Method tested K=2 to K=8 → K=4 selected
7	Run KMeans	KMeans(n_clusters=4, random_state=42)
8	Visualize	PCA reduced to 2D for cluster plot



2. Customer Segment Profiles



■ High Value Professionals

Customers: 1030 (39.2%)	Avg Age: 48 yrs	Spending: Average	Family Size: 2.9
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Middle-aged (avg 47 yrs), mostly graduated, married professionals. Average spending score. Largest segment at 1,030 customers (39%). Top profession: Artist/Engineer mix. Stable income, family-oriented.

■■■■■ Family Oriented

Customers: 596 (22.7%)	Avg Age: 27 yrs	Spending: Low	Family Size: 4.1
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Young customers (avg 27 yrs), large families (avg 4.1 members), low spending. 596 customers (23%). Mostly unmarried, not graduated. Budget-conscious buyers.

■ Young Starters

Customers: 354 (13.5%)	Avg Age: 71 yrs	Spending: Average	Family Size: 2.1
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Senior customers (avg 71 yrs), smaller families, average spending. 354 customers (13%). Mostly married, moderate graduation rate. Top profession: Lawyer. Comfortable but cautious spenders.

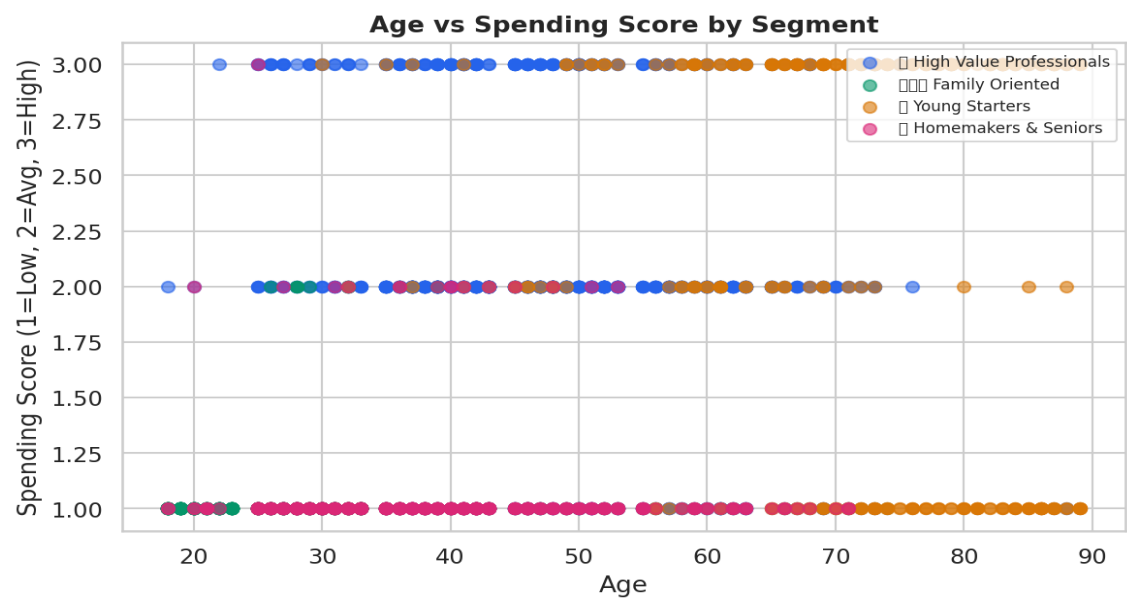
■ Homemakers & Seniors

Customers: 647 (24.6%)	Avg Age: 38 yrs	Spending: Low	Family Size: 1.8
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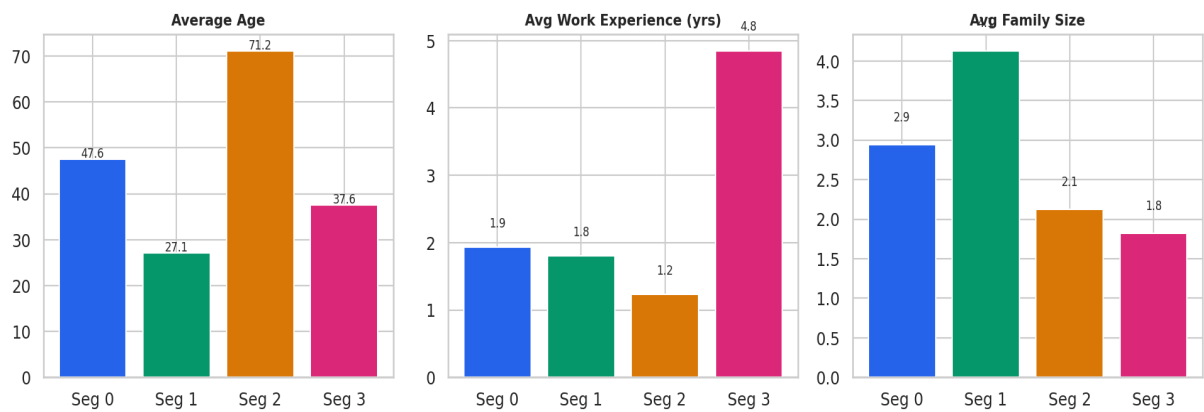
Mid-age customers (avg 37 yrs), highly graduated, good work experience (4.8 yrs). 647 customers (25%). Low spending despite education. Top profession: Artist. Career-focused, not yet high spenders.

3. Segment Visualizations

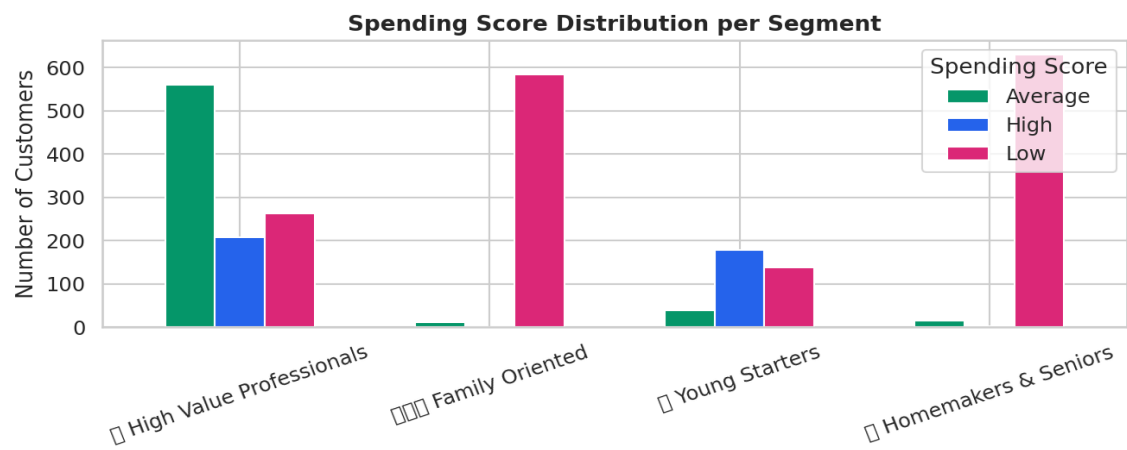
Age vs Spending Score — each dot is one customer, colored by segment:



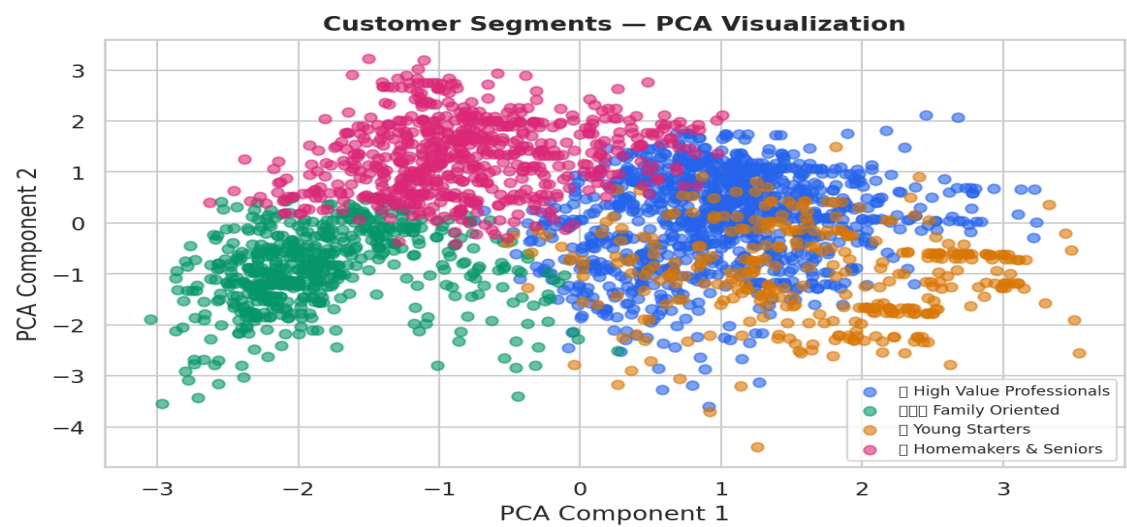
Key Metrics Comparison Across Segments:



Spending Score Distribution per Segment:



PCA 2D Cluster Visualization (confirms clear separation):



4. Targeted Marketing Recommendations

■ Strategy 1 — Premium Loyalty Program for High Value Professionals (Segment 0)

Target Segment: ■ High Value Professionals

- Launch a tiered loyalty program with exclusive benefits for top spenders.
- Offer family bundle packages — this segment has married, mid-age customers with families.
- Use email marketing with personalized product recommendations based on profession.
- Measurable KPI: Increase repeat purchase rate by 20% within 6 months.
- Channel: Email, LinkedIn ads, in-app notifications.

■ Strategy 2 — Budget-Friendly Campaigns for Family Oriented Segment (Segment 1)

Target Segment: ■■■■■ Family Oriented

- Create group discount offers — buy for the whole family, save more.
- Offer installment/EMI payment options to reduce purchase barrier.
- Run back-to-school or seasonal family offers at key times of year.
- Measurable KPI: Convert 15% of Low spenders to Average spending tier in 3 months.
- Channel: WhatsApp, Facebook ads, SMS campaigns.

■ Strategy 3 — Career & Aspiration Upsell for Homemakers & Seniors (Segment 3)

Target Segment: ■ Homemakers & Seniors

- This segment is highly educated with good work experience but low spending — untapped potential.
- Target with aspirational and career-growth products (tools, courses, premium services).
- Use limited-time upgrade offers to trigger spending urgency.
- Measurable KPI: Increase avg spending score from Low to Average within 4 months.
- Channel: Google Search ads, professional networks, email drip campaigns.

Report based on Test.csv (2,627 customers). Method: KMeans Clustering. Tool: Python (scikit-learn, pandas, matplotlib).